

Music-Gen v5 — cycles 140-141

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Abstract

This report covers the last two working cycles of the Music-Gen v5 reopening: the interpolation demonstration (feature F5) and the closing documentation (feature F7). Both landed, and the v5 arc re-closed.

The interpolation demo blends two donor songs — Chicken Grease (CG) and Peach Dream (PD) — at weight $t = 0.5$ by mixing the parameters of two donor-conditioned models (the groove tables and the harmony Markov chain), then sampling one song from the blended model with the generator’s existing SHA-256 inverse-CDF draws. No note-level averaging occurs anywhere. The demo was pre-registered before any code was written, rendered as the sixth song of generation iteration 5 (seed 4), and reproduced byte-for-byte three ways: an in-process second render, an independent second process into a fresh temporary directory, and — with the new flag absent — a re-render of iteration 4 that reproduces all five of its earlier audio files. All three pre-registered audibility clauses held on the raw (pre-normalisation) render. Verdict: **F5_LANDS**. One structural caveat is carried forward: the two donors share almost no exact groove contexts, so the groove half of the blend is a union-vocabulary mixture (each conditional row is half a donor row and half a uniform distribution over the other donor’s vocabulary), while the harmony half is a genuine mixture on 53 of 81 states.

The closing cycle appended three documents additively — a v5 section in the completion report, operator-decision entry #21, and a v5 layer in the codebase guide — with byte-equality of every pre-existing prefix asserted by test. Four parent milestones rolled up with frozen-enum verdicts (corpus landed with the tempo axis stopped; rules landed with honest gaps; ear restored as informational only; features F1–F5 landed as an honest best-of at stall 5/12), and the close milestone recorded `V5_CLOSE_LANDS` with every clause computed from disk. An independent audit re-rendered both the flag-on and flag-off runs, reproduced every hash, and found no critical or moderate defects apart from one disclosure-class pointer (a validator record split into a sibling file that no ledger event adopts). The audit’s decision was COMPLETE: no further cycle is warranted unless the operator reopens.

1. Introduction

1.1 State at the opening of this range

The v4 closure campaign closed at on-disk cycle 77 and was reopened on 2026-09-06 by operator guidance for a v5 arc (rules, ear, generation). A later operator instruction of 2026-09-09 set the rule that governs this range: finish the feature set; the ear does not need to be perfect yet. Feature landings, not the count of songs scoring ≥ 6 on the automated ear, are the gate for progress. The backlog is seven features:

Feature	Content	Status at range open
F1	length + form + arrangement	FORM_PLAN_PARTIAL (4/5 at landing)
F2	bass line + melody models with dynamics	F2_PARTIAL (RMS-variance clause unmet)
F3	guitar / piano / other comping parts	F3_LANDS
F4	tempo fix for Peach Dream and Disco A	CLOSED by operator addendum (tempi adopted)
F5	interpolation demo, two donors, $t = 0.5$	open — this range
F6	iteration schedule recorded in the stall counter	entries present since iteration 2
F7	close documents, only after F1-F5 have landed	open — this range

Four generation iterations had been rendered (seeds 0-3), each with two-fold byte-determinism proofs. The generator image at range open had SHA-256 prefix 3d0f3a24...; the stall counter stood at 4 iterations of a 12-iteration budget with zero declared passers.

1.2 Terms used throughout

- **Donor**: one of five focus songs whose transcribed MIDI conditions the generator (CG = Chicken Grease, WIG = What If I Go, Rome, PD = Peach Dream, Disco A). Songs are identified on disk by a 16-hex prefix of their SHA (CG 31a164f845f8e27e, PD 88d247468cb6d49f).
- **n=23 models**: the harmony Markov chain (81 functional chord states, relative to the tonic), the groove model (four conditional tables: kick-8th marginal, snare-16th | kick, hat-16th | kick+snare, bass-16th | kick), and the comping statistics, each fitted on the 23 eligible corpus songs. They drive the regular renders.
- **Pre-registration (“prereg”)**: a JSON file written before any code edit or output, fixing the verdict vocabulary, the acceptance clauses with tolerances, and the SHA-256 of every held-constant input. A test asserts that its modification time precedes every output.
- **Frozen enum**: a verdict must be one of the pre-declared values; nothing is retuned after seeing the result (the run’s “FD-1” rule).
- **Byte-determinism $\times 2$ / replay proof**: every render is produced twice, the second time into a fresh temporary directory; the audio files must be byte-identical (REPLAY_PROOF_HOLDS).
- **Flag-off regression**: after any additive generator edit, the previous iteration’s exact command is re-run under the new image with the new flag absent; it must reproduce the earlier audio SHAs, proving the edit is inert when off.

- **Informational ear score:** an automated VGGish-based score on a 1–7 scale. Earlier work proved that no monotone calibration of this statistic can both pass its sanity gate and discriminate the target band, so scores are recorded but never declare a passer (rule “FD-6”: the operator’s ear decides).
- **Stall counter:** the v5 generator budget — 12 iterations, target 5 passers; the counter advanced 4 → 5 in this range.
- **Cycle numbering:** the work is recorded on disk as cycles 89 and 90; the workers labelled them harness cycles 133 and 134 under an assumed offset of 44; this report’s range is harness 140–141. The offsets are disclosed, not reconciled. Throughout, “the demo cycle” means on-disk cycle 89 and “the close cycle” means on-disk cycle 90.

2. Approach

Both cycles followed the discipline established for F1–F3:

1. Write the pre-registration first (verdict enum, clauses, tolerances, held-constant SHAs, expected output counts).
2. Make the generator edit additive behind a default-off flag whose sub-flags error out without the parent flag.
3. Render; prove byte-determinism $\times 2$; run the flag-off regression against the previous iteration; score informationally; deliver SHA-verified listening copies.
4. Record the verdict mechanically from the enum; advance the stall counter with a schedule entry (F6); adopt tests; register plan rows; emit ledger events with `agent=worker`.

The close cycle rendered nothing. It appended documents additively (a prefix byte-equality test guards each pre-existing file), rolled up the parent milestones with frozen enums, triaged validator warnings, decided a housekeeping question about pipeline runners, and computed the close verdict from disk state rather than asserting it.

3. Findings

3.1 The F5 pre-registration and blend semantics

The prereg `data/v5/gen/f5_prereg_c89.json` (SHA-256 `fb614843...`) was written at 03:14:04 Z, before the generator edit, the new module, and every one of the 250 files it predicted. It fixes:

Donors and weight. A = CG (tonic F# minor, 92.285156 BPM from the corpus tempo model), B = PD (at the operator-adopted 122.197271 BPM, served from its re-canonicalised MIDI). $t = 0.5$, fixed by the plan; no sweep.

What “donor-conditioned model” means. The $n=23$ construction (same alphabet, phase alignment, $\alpha = 0.5$ smoothing, template matching, functional states relative to the donor tonic) applied to that donor’s own lossless canonical MIDI — obtained by read-only import of the existing `groove_v5_v2.load_song/table` and `harmony_v5.analyse_song` functions.

Groove blend. For each of the four conditional tables, over the union of the two donors’ outcome vocabularies:

$$P_{\text{mix}}(o \mid c) = t P_A(o \mid c) + (1 - t) P_B(o \mid c),$$

where $P_X(o \mid c)$ is donor X’s smoothed row when context c was seen in X, else uniform over X’s vocabulary (the generator’s existing `row_for` convention); an outcome absent from a donor’s vocabulary contributes 0 from that donor; contexts seen in neither donor fall back to uniform over the union. Sampling is unchanged from iteration 4: inverse-CDF on $u = \text{SHA-256 of the}$

same per-bar tags. The n=23 corpus tables are not consulted for the demo groove.

Harmony blend. Over the n=23 chain’s 81 functional states, with $n_X(s)$ = donor X’s segment count leaving state s :

$$\text{row}_{\text{mix}}(s) = t P_A(s, \cdot) + (1 - t) P_B(s, \cdot), \quad n_{\text{mix}}(s) = t n_A(s) + (1 - t) n_B(s),$$

where $P_X(s, \cdot)$ is donor X’s row-normalised segment transitions if $n_X(s) > 0$, else the n=23 chain’s row (a back-off, counted and disclosed per state). The blended chain stores $n_{\text{mix}}(s) \cdot \text{row}_{\text{mix}}(s)$ as segment-level counts, so the generator’s existing matrix builder recovers the mixed row and the F1 section-contrast rule (≥ 8 segments) reads the blended support. Start weights $\pi_{\text{mix}} = t \pi_A + (1 - t) \pi_B$. The tonic is donor A’s (states are functional, so the blend is key-independent; a tonic cannot be averaged).

Declared deviation. The research brief had recommended argmax decoding with a tiebreak. The prereg declines it: argmax decoding of a Markov chain collapses to a fixed one-or-two-state cycle and is not a chord progression. Decoding stays the SHA-256 inverse-CDF used by every other v5 chord draw. The prereg governs.

What is not blended. Tempo, tonic, instrument profiles, F2 velocity/bass/melody models, and F3 comping are donor A’s or the standard n=23 assets. No arithmetic mean of two rendered note streams or of two sampled sequences is performed anywhere (the ban inherited from the v4 interpolation demo).

Verdict enum and clauses. F5_LANDS iff (demo byte-determinism in-process AND in an independent process) AND replay proof $\times 2$ AND flag-off regression 5/5 AND every audibility clause; F5_PARTIAL iff determinism and regression pass but an audibility clause fails; else F5_FAILS. The audibility clauses are measured on the raw render, before RMS normalisation (closing a prior audit’s note that a post-normalisation check is tautological):

- (i) every demo stem’s raw render RMS > -60 dBFS;
- (ii) drum density: with $d(t)$ = mean drum onsets per composed bar, $|d(0.5) - \frac{1}{2}(d(1) + d(0))| \leq 2.0$, where $d(1)$ and $d(0)$ come from re-composing the same tags with the A-only and B-only tables;
- (iii) chord Hamming distance to each donor-only sequence: $0 < h_A, h_B < 1$ and $|h_A - h_B| \leq 0.35$ (distinct from both donors, not lopsided).

3.2 Generator wiring and a shared repo-root helper

scripts/v5/generate_v5.py was edited additively (SHA 3d0f3a24... $\rightarrow$ 94c8ba99...): --f5 default off, with sub-flags --interp-a, --interp-b, --interp-t, --f5-prereg that raise an argparse error without --f5; --f5 itself requires all four plus --f2 --f3 --form-plan. With the flag on, the five regular songs render exactly as before and one demo specification is appended with the blended models. The manifest gains an f5 block: raw render RMS of every stem, the density and Hamming statistics against A-only and B-only compositions on the same tags (via a new f5_compose mirror of the section composition, asserted equal to the demo’s own composition), and the blend record.

Two new modules: scripts/v5/interpolate_v5.py (190 lines, SHA 177ac5c6...; imported only under --f5; ships a self-check on a synthetic donor pair) and scripts/v5/repo_root.py (42 lines, SHA d012e510...), a marker walk-up that lets plot scripts under data/ locate the workspace without a hard-coded parents[N] — closing a recurring housekeeping note from earlier audits.

One in-cycle slip was disclosed: the smoke render named the demo t_{500} ($t \times 1000$); it was corrected to t_{050} ($t \times 100$, the prereg identifier) before the real launch, and the smoke directory was pruned.

3.3 Iteration 5: five regular songs and the demo

The iteration-5 command (seed 4; --f2 --velocity-mode f2 --f3 --tempo-overrides ... --f5 --interp-a CG --interp-b PD --interp-t 0.5 --prove-replay) was launched detached and rendered six songs. Every one has form AABABCAA over 64 bars with the bass-only intro and breakdown at section 2.

Song	Donor	Tempo (BPM)	Duration (s)	ab_mix.wav SHA-256
song 1	CG	92.285156	173.7099	eae40e28...
song 2	WIG	99.384014	159.6111	c9b65ce9...
song 3	Rome	151.999081	106.0600	9d91390a...
song 4	PD	122.197271 (adopted)	130.5527	28e18c11...
song 5	Disco A	120.272335 (adopted)	132.6498	45f10a8a...
demo gen_v5_ interp_CG_PD_t050	A = CG, B = PD	92.285156 (donor A)	171.4808	947b348a...

Regular-song feature enums were unchanged from iteration 4: F1 FORM_PLAN_LANDS 5/5, F2 F2_PARTIAL (the RMS-variance clause was not requested this iteration), F3 per-song clauses 5/5 (every comping part ≥ 32 note-ons and audible; guitar 2090–2305, piano 1500–1770, other 2580–2766 note-ons per song).

Demo statistics (from the pinned rollup and byte-determinism record):

Quantity	Value	Clause
raw render RMS, dBFS — bass / drums / guitar / keys / melody / other / piano	−31.86 / −30.91 / −21.16 / −23.01 / −30.50 / −18.34 / −24.22	(i) all > −60: holds
drum density $d(0.5)$ vs $d(1)$ (A-only), $d(0)$ (B-only)	11.890625 vs 13.0, 12.0; expected mixture 12.5; deviation 0.609	(ii) ≤ 2.0 : holds
chord Hamming h_A, h_B , $ h_A - h_B $	0.9375, 0.8125, 0.125	(iii) in $(0, 1)$ and ≤ 0.35 : holds
bass-kick lock (informational) — mix / A-only / B-only	0.594 / 0.790 / 0.392	none
tonic	6 (F# minor, donor A)	—
F3 parts, note-ons — guitar / piano / other	2160 / 1644 / 2514	≥ 32 each

One bookkeeping inconsistency is noted, not resolved: the cycle-close narrative in the ledger quotes the demo density as 13.6 per bar “above both donors’ 12.4 / 11.6”, whereas the pinned rollup, the byte-determinism record and the figure all carry 11.89 vs 13.0 / 12.0. The pinned

records are what the tests and the audit checked; the narrative numbers have no on-disk source.

Blend support (why the caveat in §4 exists). Groove: each of the four tables has exactly one context seen in both donors (the unconditioned kick marginal); the three conditional tables have 55/14, 73/29 and 55/14 A-only/B-only contexts and zero shared. Vocabulary sizes A/B/union: kick 56/15/70, snare 22/9/30, hat 26/25/49, bass 88/34/121. Harmony: donor A contributed 280 segments and B 164; states with donor support A 41, B 27, blended 53 of 81; $n=23$ back-off rows used for 12 states on the A side and 26 on the B side; 28 states unseen in both keep the uniform-row convention. The demo’s section-contrast rule found only 6 allowed start states for B and C sections (the regular songs have 51–55), a direct consequence of the thinner blended support.

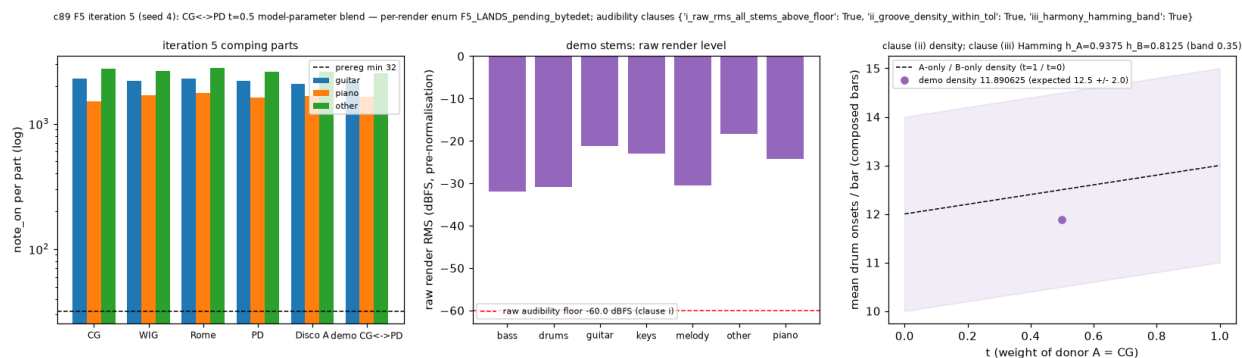


Figure 1: Iteration 5 (seed 4). Left: comping-part note-on counts for the five regular songs and the demo, all far above the pre-registered minimum of 32 (dashed). Middle: raw, pre-normalisation render RMS of every demo stem against the -60 dBFS audibility floor (clause i). Right: demo drum density (dot) at $t = 0.5$ against the line joining the B-only ($t = 0$) and A-only ($t = 1$) densities, with the ± 2.0 onsets/bar tolerance band (clause ii); the title carries the Hamming statistics for clause iii.

3.4 Determinism three ways, and the verdict

Check	Result	Wall time
In-process replay (<code>--prove-replay</code> , second render into a fresh temp dir)	6/6 REPLAY_PROOF_HOLDS	—
Independent second process (exact command, fresh mkdtemp, <code>--no-stall-update</code>)	6/6 equal on <code>ab_mix.wav</code> , every per-stem MIDI, every per-track file, and <code>f5_blend.json</code>	188.1 s
Flag-off regression: iteration-4 command (seed 3, no <code>--f5</code>) under the post-edit image	5/5 iteration-4 SHAs reproduced (a501df3c..., fe366926..., 19b0f51d..., 99e3ee69..., 313b98bd...); iteration-4 files on disk byte-identical	151.9 s
Informational score table, two runs	identical (SHA e8a39319...)	—

With every clause true, the byte-determinism record folded in `f5_enum_final = F5_LANDS`, and the ledger recorded the verdict, discharging both the operator’s long-owed “decision #7” (a

v5 interpolation demo) and an earlier audit note that F5 needed its own prereg.

3.5 Ear scores, stall counter, delivery

Informational scores (VGGish, the c76 “wider-linear” calibration through the isolated venv), all ≥ 6 and all inside the 5.70–6.72 band-4 context range that the earlier infeasibility proof showed the statistic cannot separate:

CG	WIG	Rome	PD	Disco A	demo
6.4615	6.4201	6.2187	6.5625	6.4967	6.4085

No passer was declared. The stall counter advanced to **5/12** with the F6 schedule entry (iteration 5, seed 4, feature “F5 interpolation demo”, donors CG/PD, $t = 0.5$, the form / harmony / groove / comping / prereg / blend-record SHAs, verdict F5_LANDS). Listening copies of all six renders went to data/v4/generated/v5_iter_05/ and the demo trio (audio, manifest, replay proof) plus f5_blend.json and a copy of the prereg to data/v4/generated/f5_interp_CG_PD_t050_c89/, each SHA-verified by a manifest.

3.6 Housekeeping and tests in the demo cycle

- Disk: 86 % used at open; 565 MB of regenerable prior-session render temp directories pruned under /tmp; 84 % after; never ≥ 90 %.
- Tests: tests/test_c89_landing.py (11/11) — prereg mtime gate and not-future-dated stamp, flag guards, byte-determinism $\times 2$ plus the independent process, F5-off reproduces iteration 4, pre/post generator SHAs, audibility statistics recomputed from the render within the prereg bands, blend record and composition reproduced from a fresh build, expected-output count equal to disk including the figure, discipline scan with no parents[N], stall 5/12 and delivery SHAs, verdict/plan-row consistency, iteration 1–4 audio and read-only pins byte-identical. Generator-SHA pins in the c86–c88 suites were re-pinned to the chain (c88 post-edit == c89 pre-edit), and one stall pin was loosened from == 4 to ≥ 4 because the counter advances by design; both re-pins are commented in place. Adopted suite: 11 files, 84 PASS lines, all exit 0.
- Optional $n=25$ sibling models (the $n=23$ set plus the two late-landing songs 0e1e8f20..., cc0693b4...) were skipped for the second time; nothing consumes them.
- Ledger 2138 $\rightarrow$ 2152 (14 events); plan rows registered; the pipeline runners lived only in the session scratchpad (their commands are pinned in the launch JSON and byte-determinism record) — a gap the close cycle addressed.

3.7 The close cycle: three documents appended additively

With F1–F5 landed, the F7 clause of the backlog guidance opened. Three documents were appended, each below its existing content; a test asserts `sha256(current[:pre_bytes]) == pre_sha256` for every file:

Document	Before → after	What was added
docs/v4_completion_report_v3.md	20 497 → 54 331 B (b900b0ee... → 8341e526...; first 8 KB and title tag unchanged)	“Section: v5 REOPENING (c79-c90)”: cycle-counter and outage disclosure; per-milestone narratives for corpus, rules, ear, generation; the F1-F7 table with landing-cycle verdicts; a 26-row render table; the F5 caveat verbatim; carry-forward hygiene; program coverage; determinism receipts and generator lineage; the validator table; the runners decision; a deliverable index of 109 path/SHA pairs (every one resolving on disk); honest gaps; close rationale
docs/OPERATOR_DECISIONS.md	16 147 → 19 655 B	Entry #21 “v5 CLEAN RE-CLOSE at c90”: verdict matrix, F1-F7 table, the binding F5 caveat, the ear rule, delegation of the 26 v5 renders and 25 v4 A/Bs to the operator’s ear, tempo-axis STOP, re-close wording with no finality claim
docs/CODEBASE_GUIDE.md	4 536 → 12 997 B	“v5 REOPENING layer”: module map with 21 SHAs, generator flag matrix, data/v5/ layout, v5-specific disciplines, blend_record_sha256 semantics

Every number in the v5 section was read from disk at write time; the audit later re-derived each one from the source JSON (harmony max stationary mass 0.069317, groove singleton-context fraction 0.637448, comping pooled max slot mass 0.065174, melody VOMM order-3 singleton 0.720574, the ear-gate values, the demo gains and raw levels, the F3 note-on counts, all 26 informational scores) and found them consistent, with two prose slips noted in §3.10.

3.8 Parent roll-ups and the close verdict

Four parent milestones rolled up with frozen enums, each restating recorded verdicts without re-measuring anything:

Milestone	Enum	Substance
corpus	CORPUS_LANDED_TEMPO_AXIS_STOPPED	26 songs transcribed full-length and reindexed lossless with sidecars; the index-collision defect fixed; 1 song content-blocked; four tempo criteria each falsified under its own prereg; PD 122.197271 / Disco A 120.272335 BPM by operator adoption
rules	RULES_LANDED_WITH_HONEST_GAPS	harmony n=23 non-degenerate; groove overfits (0.637); comping non-degenerate but slot histogram structureless; form model failed its test so a fixed template governs; velocity profiles structureless; melody VOMM memorises (0.72)
ear	EAR_RESTORED_INFORMATIONAL_ONLY	isolated venv rebuilt and amended; leave-one-out gate $5/5 \geq 6$ (min 6.2095); band-4 spot check fails (6.7199 > 5.7095); monotone-infeasibility proof stands
generation	FEATURES_F1_F5_LANDED_HONEST_BEST_OF_AT_STALL_5_OF_12	F1 partial, F2 partial, F3 lands, F4 closed by operator, F5 lands; F6 entries complete; stall 5/12, passers 0 by design; a sixth iteration is not opened (it would be preservation-spin)

The close milestone then recorded **v5_CLOSE_LANDS** from a check dictionary computed by the emitter from disk: no anchor drift (44 turn-start pins; the only four that moved were the three appended docs and the plan file), docs prefix byte-equal, v3 first 8 KB unchanged, validator ERROR count at baseline, tests green, no render trees touched. The event’s wording makes no finality claim: the run re-closes cleanly; the operator verifies post-close; a later guidance file may reopen exactly as the v5 reopening did.

3.9 Validators, the adopt event, and the runners decision

Validator triage. The project’s promise checker reported 156 errors / 8 960 warnings at open. The 156 errors are all pre-existing (98 events missing an agent field from an old emitter chain, 40 unregistered milestone ids, 16 out-of-vocabulary statuses, 2 illegal transitions; zero on any line from these cycles). The warning growth (+253 since the previous cycle) was entirely the “orphan artifact in managed path” class fired on the new iteration-5 tree. Since

the emitter can carry ≥ 250 paths, a single adopt event registered the iteration-5 tree, the demo delivery and the listening copies (280 paths) under the generation milestone, reducing warnings to 8 731; after the closing events the count was 8 714, then 8 707 after a byte-exact restore of the frozen record, then 8 708 after the record split described next. The organisation checker stayed at 0 errors / 54 warnings (pre-existing workspace-root files).

The warning series c85 $\rightarrow$ c90: 8 543, not recorded, not recorded, 8 707, 8 960, 8 960 (open) $\rightarrow$ 8 731 (after adopt).

Record split. data/v5/logs/validators_c90.json is quoted by SHA in the completion-report deliverable index, so it was frozen at its after-adopt state and the later final / post_close phases were written to a sibling validators_c90_close.json. No ledger event adopts the sibling; the v5 section still points at the original file's final phase. This is the audit's one moderate finding (§3.10).

Runners decision. A prior audit had noted that the per-cycle pipeline runners (prune $\rightarrow$ pins/prereg $\rightarrow$ smoke $\rightarrow$ detached launch $\rightarrow$ independent process $\rightarrow$ flag-off $\rightarrow$ score $\times 2 \rightarrow$ deliver $\rightarrow$ figure $\rightarrow$ byte-det assembly $\rightarrow$ tests) existed only in a session scratchpad. Option (a) was chosen: check in scripts/v5/runners/run_from_launch_json.py — a generic re-executor that prints (default, dry-run) or launches detached the exact command pinned in any launch JSON or byte-determinism entry, refuses to execute on a drifted generator image or without a fresh --out, and reproduces the pinned strings byte-for-byte (asserted by test) — plus a README and verbatim copies of the ten c89 runners under scripts/v5/runners/c89/ as a historical record. Rationale recorded: every command was already pinned, so the cost is one small module and the benefit is a re-run that needs no retyping and refuses drifted images.

Other close-cycle housekeeping. Disk 87 % $\rightarrow$ 85 % after pruning 596 MB of regenerable render temp directories (this session's and, listed by path and size before deletion, the previous auditor's two re-render directories). tests/test_c90_close.py (9/9) plus the c84-c89 suites re-run unchanged: 12 files, 93 PASS lines, all exit 0. Ledger 2152 $\rightarrow$ 2171 (19 events); plan rows for the close milestones amended additively. The n=25 sibling models were skipped a third time and recorded as an honest gap rather than deferred again.

3.10 The independent audit of the close cycle

The auditor verified by execution, not by reading claims: prefix byte-equality and post-SHAs of the three docs; all 109 SHA pairs in the deliverable index and 21 in the module map resolving byte-exactly; all 26 render rows matching their rollups; every quoted number against its source JSON; 40 of 44 turn-start pins byte-identical with the four movers accounted for; the generator unedited; no file under any render tree newer than cycle open; all 19 close-cycle ledger events well-formed with every listed artifact present; validators recomputed at 156 / 8 708 and 0 / 54; 12 test files executed (105 PASS lines, all exit 0); the runner's dry-run reproduced both pinned commands byte-for-byte and refused --execute without --out. The auditor had independently re-rendered iteration 5 (flag on) and iteration 4 (flag off) in the previous audit, reproducing every SHA.

Findings:

- Critical: none.
- Moderate (one, disclosure-class, deliberately not patched): the validator-record split (§3.9). The minimal fix — an auditor-authored adopt event — would break a test asserting every close-cycle event has agent=worker and hence the tests_green clause of the close verdict; recorded as a recommendation instead.
- Minor (nine, noted): the v5 section's error-composition prose says "42 unregistered ids + 2 transitions" (sums to 158) where disk shows 40 + 2 (the total 156 is correct and test-

asserted); the comping IOI figure “47.5 % at one 16th” is the guitar-stem value, pooled is 44.7 % (qualitative claim unaffected); the close narrative lacks an explicit ledger-count and tests line; the sibling validator file is an orphan (+1 warning); the runner’s created stamp predates its file mtime (not future-dated); 6 830 orphan-artifact warnings remain, mostly pre-v5 sweep renders; several v5 files show as modified or untracked versus the last commit (commit lag; live SHAs equal the pins); the n=25 siblings were never built; a scratch-archive event names its builder scripts differently from the actual builder.

Decision: **COMPLETE**. The directive’s “drive itself to a clean close” was satisfied at cycle 77 and is re-satisfied at cycle 90 after the operator’s reopening; the operator’s rule makes feature landings the gate, and F1-F5 are all landed with their landing-cycle enums. The residuals are operator-side (listening) or too small for a cycle that would not be a null cycle.

4. Discussion

What the demo demonstrates and what it does not. The F5 result establishes that the v5 generator can consume a blended model built by a documented, pre-registered rule and produce a deterministic, audible, structurally valid song that differs from both donor-only compositions (Hamming 0.94 and 0.81) while sitting within tolerance of the mixture density. It does not establish that the blend is musically “between” the donors: with zero shared conditional groove contexts, half of every conditional row is uniform noise over the other donor’s vocabulary, and the harmony blend leaned on n=23 back-off for a third of its states. Operator-decision entry #21 makes the remedy binding for any follow-up: a *t*-sweep or a second donor pair must first pre-declare shared-context conditioning (coarser contexts) or the per-position SHA-256 fallback used by the v4 demo.

Determinism held under a fifth generator image. Iterations 1-5 were produced under successive additive edits (3fbd98ca → b3473fd2 → f261690c → 3d0f3a24 → 94c8ba99), and each edit reproduced the previous iteration with its flag off. The independent-process check added this range removes in-process state as a possible explanation for replay agreement. The environment pin has not changed since on-disk cycle 22.

Where the small slips live. The audit’s two number slips (error composition, IOI percentage) are both in prose, not in SHA tables; every hash-checked value reproduced. The same pattern shows in the density numbers quoted in the demo cycle’s closing narrative versus the pinned rollup (§3.3). The audit’s recommendation — generate prose counts from the same JSON the tests read — follows directly.

Honest gaps carried out of the arc. The tempo axis stopped after four falsified criteria, so PD and Disco A tempi rest on operator adoption. The groove model overfits, the comping slot histogram is structureless, the form model failed its test, and the melody model memorises. F2’s RMS-variance clause remains unmet. Fourteen of fifteen (song, comping-part) cells render through General MIDI shims. The demo’s bass and drum normalisation gains hit the 4.0 cap, leaving those stems 1-2 dB under target with no clause governing it. The automated ear cannot discriminate this content; no operator listening verdict exists yet on the 26 v5 renders or the 25 v4 A/Bs.

5. Open questions and next steps

No further cycle was scheduled. If the operator reopens:

1. First act: one worker event adopting data/v5/logs/validators_c90_close.json and a one-line additive correction that the final / post_close validator phases live in that sibling file; fold in the two prose corrections (40 unregistered ids, not 42; pooled IOI 44.7 %, guitar

stem 47.5 %).

2. Any *t*-sweep or second donor pair must pre-declare shared-context conditioning or the per-position fallback before rendering (binding).
3. The $n=25$ sibling models remain optional and non-consumed.
4. The 26 v5 listening copies and the demo delivery await the operator's ear; no passer can be declared by the automated score.
5. Commit lag: generate_v5.py, MANIFEST.md, the velocity-aware serializer and the scripts/v5/{comping_*, interpolate_v5, repo_root, runners/} files are uncommitted; live SHAs equal the pinned ones.

Appendix: Implementation Details

Code organisation (this range)

```
scripts/v5/
  generate_v5.py          1114 lines  (+--f5 and sub-flags; sha 3d0f3a24... -> 94c8ba99...)
  interpolate_v5.py      190 lines   new (sha 177ac5c6...)
  repo_root.py           42 lines   new (sha d012e510...)
  runners/
    run_from_launch_json.py 110 lines  new (sha 9c6f464e...)
    README.md               new (sha 10c72355...)
    c89/*.py                10 files, 413 lines, verbatim copies of the demo-cycle runners
data/v5/gen/
  f5_prereg_c89.json      sha fb614843... (03:14:04Z, precedes all outputs)
  byte_determinism_c89.json sha 476269f3...
  byte_determinism_c90.json turn-start pins for the close cycle
  gen_v5_iter05_ear_scores_c89.json
  stall_counter.json      sha 4b87958f... (5/12)
  plot_iter05_parts_c89.py 83 lines -> fig_iter05_parts_c89.png (sha 2668bed3...)
  iteration_05/           248 files: 5 regular renders + demo (+ f5_blend.json), rollup sha 4787a346...
data/v4/generated/v5_iter_05/ 6 listening copies + listening_manifest.json (sha d8647e7a...)
data/v4/generated/f5_interp_CG_PD_t050_c89/ demo delivery, delivery_manifest.json (sha 7cddb63a...)
data/v5/logs/
  gen_iter05_c89.launch.json, gen_iter05_c89.log, indep_c89.log, flagoff_c89.log, bytedet_c89.log,
  deliver_*_c89.log, score_gen_v5_iter05_c89_run{1,2}.log, c89_prune.json, c90_prune.json,
  validators_c90.json (sha 9f0db6e2...), validators_c90_close.json, runners_decision_c90.json,
  test_results_c89.json, test_results_c90.json
docs/
  v4_completion_report_v3.md +v5 section (20497 -> 54331 B; 8341e526...)
  OPERATOR_DECISIONS.md     +entry #21 (16147 -> 19655 B; 9599b31e...)
  CODEBASE_GUIDE.md         +v5 layer (4536 -> 12997 B; 4d08a0b5...)
tools/
  _emit_c89_ledger_events.py 229, _register_c89_por_rows.py 115,
  _emit_c90_ledger_events.py 354, _register_c90_por_rows.py 110
tests/
  test_c89_landing.py 286 lines, 11 tests; test_c90_close.py 223 lines, 9 tests
```

Test results

- Demo cycle close: 11 files / 84 PASS lines / all rc 0 (test_results_c89.json); test_c89_landing.py 11/11.
- Close cycle: 12 files / 93 PASS lines / all rc 0 (test_results_c90.json); test_c90_close.py 9/9;

no re-pins.

- Auditor re-execution (close cycle): 12 files, 105 PASS lines, all rc 0; runner dry-runs byte-for-byte; --execute without --out refused (rc 3).

Key hashes

Item	SHA-256
generator (current)	94c8ba99eac98eee1f88d871cfc520507ddf253d3962287f50a7a368fdaaa
F5 prereg	fb6148435c15b8bbff64dc13e4b3d8cd5501f00ab182b4de757dd9897dd2c
demo ab_mix.wav	947b348a10ab8c5c89491cf1a88eeabda680dbbb4f7d8620899a846ec0d7d
blend record (compact canonical JSON) / f5_blend.json file bytes	0bc6f353... / 72d9cdcd...
blended groove model / blended harmony chain	dcf0af85... / bab58797...
n=23 harmony chain / groove / comping / tempo overrides	330b9d46... / 57072025... / 01024254... / ef52f2a0...
env pin (unchanged c22 → c90)	2ac444c36298d6ada0579abala9160a5881703a4e628f5cccd828b842a92

Ledger and plan

- Ledger lines 2139-2152: demo cycle (14 events, cycle=89): _plan/guidance-continuation-c89, _infra/disk-prune-c89, M-V5-GEN-1/f5-prereg-written-c89, _infra/f5-wiring-and-repo-root-helper-c89, M-V5-GEN-1/f5-iter05-rendered-c89, M-V5-GEN-1/f5-flagoff-regression-c89, M-V5-GEN-1/f5-verdict-c89, M-V5-GEN-1/F5-interpolation-demo, M-V5-GEN-1/stall-counter-5of12-c89, _infra/n25-siblings-deferred-c89, _plan/register-c89-sub-leaves, _infra/adopt-cycle89-tests, _archive/cycle-89-scratch, _run/cycle_89_closed.
- Ledger lines 2153-2171: close cycle (19 events, cycle=90): _infra/adopt-cycle89-gen-artifacts-c90 (280 paths), _plan/guidance-continuation-c90, _infra/disk-prune-c90, _infra/warn-growth-triage-c90, _infra/runners-decision-c90, M-V5-CLOSE-1/completion-report-v5-section-c90, _plan/operator-decisions-c90-amendment, _plan/codebase-guide-c90-amendment, M-V5-CORPUS-1, M-V5-RULES-1, M-V5-EAR-1, M-V5-GEN-1, M-V5-GEN-1/F7-close, _infra/n25-siblings-skipped-c90, _plan/register-c90-sub-leaves, M-V5-CLOSE-1 (= V5_CLOSE_LANDS), _infra/adopt-cycle90-tests, _archive/cycle-90-scratch, _run/cycle_90_closed (str-supersedes _run/cycle_89_closed).
- Plan of record: F5 row M-V5-GEN-1/F5-interpolation-demo amended additively (c89 LANDED); M-V5-CLOSE-1 and M-V5-GEN-1/F7-close rows amended additively (c90 CLOSED); sub-leaves for both cycles.
- Governing guidance: guidance_2026-09-09_c131_finish_F2_iter3_then_F3.txt (sha16 fed27e550e9f7c75) and guidance_2026-09-09_finish_features_backlog_F1-F7.txt (sha16 cf7296cfcf70c277); no new guidance file landed in either cycle.

Session references

- Cycle 140: researcher 43b6ba46-bc47-4e93-9418-9f0f8431736a, worker 4bd561b5-200a-43b8-9b6e-ac7c19f31e87, auditor 9281e1ba-d955-4c05-9410-26cf3b378faa.
- Cycle 141: researcher ff7559ca-8996-4869-a5ad-aa534d28cd36, worker 694cc9a7-cecd-40ea-94aa-fb5e6c46257d, auditor ba8588cd-00e8-45ac-a4e6-1b362ca2bd04.
- Mapping used: cycle 140 ≈ on-disk c89 (workers' label harness c133), cycle 141 ≈ on-disk c90 (harness c134); disclosed, not reconciled.

Gaps in the record

- Session-history retrieval was not permitted in this reporting session; all material was gathered from disk (ledger, records, docs, logs, tests, figure) plus the supplied close-cycle audit. Researcher briefs and the demo-cycle audit report are not on disk; the demo-cycle audit's findings are known only as folded into the close documents and the close-cycle events (runners gap, prune listing, blend-record hash semantics, inherited degeneracy verdict, 4.0 gain cap).
- REFERENCES.md does not exist in the workspace; no References section is emitted.
- The demo-cycle closing narrative's density numbers (13.6 / 12.4 / 11.6) differ from the pinned rollup (11.89 / 13.0 / 12.0); the source of the narrative values was not found.
- MANIFEST.md was rewritten for this range (Key Files section preserved verbatim); no git commit was made.